

The Layered Internet Model

Lecture Overview

- Digital services are built on multiple layers
- Understanding layers helps explain **how value is created and delivered**
- Each layer depends on the previous one

“Your favorite app doesn’t just appear — it sits on a complex stack of technology.”

What apps do we use daily?

- eSewa
 - Pathao
 - Foodmandu
 - Instagram
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- These apps rely on hidden technology stacks
 - Users only see the top layer

Why the Layered Model Matters

- Shows where value is created
- Explains platform power
- Helps analyze digital competition
- Important for startups and policymakers

Nepali Context:

- Most Nepali startups operate at the application layer

Layers of Internet Model

Infrastructure Layer

- “The infrastructure layer is the **foundation** of the digital world.”

It includes:

- Physical hardware (servers, routers)
- Data centers
- Internet connectivity (4G, 5G, fiber)

Analogy:

“This is like roads and electricity for digital services.”

Layers of Internet Model - Infrastructure Layer

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Examples

- **4G/5G Networks:** Enable mobile internet
- **AWS Data Centers:** Store and process massive data

“Netflix does not run from someone’s laptop. It runs on global data centers connected by high-speed networks.”

- **Nepali Context: Ncell / NTC 4G:** Backbone of Nepal’s mobile internet
- **Bank backend servers:** Where transaction data is processed

“When Ncell or NTC network is slow, apps like eSewa or Pathao also feel slow — even if the app itself is well-designed.”

Layers of Internet Model - Platform Layer

“The platform layer sits on top of infrastructure and provides the **tools to build and run apps.**”

Includes:

- Operating systems (Android, iOS)
- Cloud platforms (AWS, Google Cloud)
- Databases, APIs, development frameworks

“Most Nepali companies do not build their own servers or operating systems. They rent or use platforms.”

Examples

- **AWS:** Provides computing, storage, databases
- **Android / iOS:** Allow apps to run on smartphones
- “If Google changes Android rules, millions of apps are affected overnight.”
- **Nabil Bank app backend:** Runs on cloud platforms
- **Android OS:** Most Nepali users use Android phones

“Nepali apps depend heavily on global platforms they do not control.”

Layers of Internet Model - Application Layer

“This is the layer users interact with directly.”

Includes:

- Mobile apps
- Websites
- Digital services

“When people say ‘I use eSewa’, they are actually referring only to the application layer.”

Examples

- **Netflix**
- **Spotify**
- **WhatsApp**

“These apps create value by offering convenience, content, and communication.”

- **Nabil Bank app:** Digital banking
- **eSewa:** Digital payments
- **Pathao:** Ride-hailing and food delivery
- **Foodmandu:** Online food ordering

“Apps compete on user experience, speed, trust, and convenience.”

Connecting the Layers

“These layers do not work independently.”

Infrastructure → Platform → Application → User

- Apps depend on platforms
- Platforms depend on infrastructure
- Weak lower layers limit innovation at the top

Nepal's Digital Reality

- Infrastructure/Networks: NTC, Ncell
 - Platforms: Android, iOS, Cloud
 - Apps: eSewa, Daraz, Pathao
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- Nepal does not control lower layers
 - This affects innovation and cost

Analogy

- Roads = Infrastructure
 - Vehicles = Platform
 - Passengers = Applications
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- Good apps fail on poor networks
 - Infrastructure enables everything

Layers of Internet Model

Layer	Description	Global Example	Nepali Example
Infrastructure Layer	Physical hardware, servers, data centers, network connectivity (4G/5G, fiber)	4G/5G networks, AWS data centers	Ncell/NTC 4G networks, Nabil Bank backend servers
Platform Layer	Operating systems, cloud platforms, development frameworks	AWS (cloud), Android/iOS (OS)	Nabil Bank app backend on AWS/Cloud, Android OS for mobile
Application Layer	Software apps used by consumers	Netflix, Spotify, WhatsApp	Nabil Bank app, eSewa, Pathao, Foodmandu

What Do We Mean by ‘Value Flow’?

- “Value flow means how **usefulness, functionality, and economic value** move from infrastructure to users.”
- Infrastructure creates **technical value**
- Platform transforms it into **usable capability**
- Applications convert it into **consumer value**
- “The lower layers create capability; the upper layers create experience.”

Layer's Role in Value Creation

Infrastructure Layer → Provides Connectivity & Computing Power

“Infrastructure does not directly serve users, but it makes everything else possible.”

What value it provides:

- Internet connectivity
- Speed and reliability
- Data processing power

Nepali Context:

- NTC / Ncell 4G networks
- Local and cloud-based data centers

“If internet speed is poor in rural Nepal, digital services cannot deliver full value, no matter how good the app is.”

Layer's Role in Value Creation

Platform Layer → Provides Tools & Foundations

“Platforms take raw infrastructure and turn it into something developers can actually use.”

Value created:

- Operating systems (Android, iOS)
- Cloud storage
- APIs and databases

Nepali Context:

- Bank apps hosted on cloud platforms
- Android OS dominating smartphone usage

Cause–Effect:

“If Android changes its policies, Nepali apps must adapt immediately — even though they don't control those platforms.”

Layer's Role in Value Creation

APPLICATION LAYER -> This is where users finally experience value.”

Value delivered:

- Convenience
- Speed
- Trust
- User experience

Nepali examples:

- Nabil Bank app → easy money transfers
- eSewa → digital payments
- Pathao → mobility and delivery

“Users pay for the application layer, but that value is built on invisible layers below.”

Example

- **4G Network (Infrastructure)**
 - Provides internet access
 - Enables data transmission
- **Nabil Bank Backend (Platform)**
 - Processes transactions
 - Stores customer data securely
- **Mobile App (Application)**
 - User transfers money
 - User checks balance instantly

“If the 4G network fails, the bank app cannot function — even though the bank system itself may be working.”

Example

- **AWS Servers (Infrastructure)**

- Provide computing and storage

- **Netflix Cloud Platform (Platform)**

- Manages content delivery
- Handles recommendations

- **Netflix App (Application)**

- Users stream movies smoothly

“Netflix’s success depends as much on cloud infrastructure as on content.”

Power Insight: Where Control Lies

“Control over lower layers often means power over higher layers.”

Examples:

- Telecom companies influence app performance
- Platform owners influence app visibility and pricing

Nepal Scenario:

- “Since Nepal does not control global platforms, Nepali startups are structurally dependent.”

Value Flow Across Layers

*“Without infrastructure, platforms fail.
Without platforms, apps cannot serve users.
Value always flows from bottom to top.”*

Benefits of Layered Architecture

- **Scalability:** Apps can grow without changing underlying infrastructure
- **Flexibility:** Platforms can host multiple apps
- **Innovation:** Developers focus on apps, not infrastructure
- **Interoperability:** Standard layers allow apps to work across devices
- **Nepali Context:**
 - Daraz can run on the same cloud infrastructure as multiple other apps
 - Banks can offer digital services without building their own network

Examples — Global vs Nepali

Layer	Global Example	Nepali Example
Infrastructure	AWS, Google Cloud, Fiber networks	NTC/Ncell 4G, Cloud hosting for apps
Platform	AWS cloud services, iOS/Android OS	Nabil Bank app backend, Mobile OS
Application	Netflix, Spotify	Nabil Bank app, eSewa, Foodmandu, Pathao